

### Advanced (Habanero)

- Q1.** What is the slope of the line  $y = 2x + 3$ ?  
(A) 2  
(B)  $-2$   
(C) 0  
(D) 3  
(E) Undefined
- Q2.** The graph of  $y = mx + b$  has a slope of  $-1/2$ . What is  $m$ ?  
(A)  $-1$   
(B)  $-1/2$   
(C) 0  
(D)  $1/2$   
(E) 1
- Q3.** Find the slope of the line passing through points  $(1, 2)$  and  $(3, 4)$ .  
(A) 0  
(B) 1  
(C)  $1/2$   
(D) 2  
(E) Undefined
- Q4.** A line has a slope of 0. What is its rate of change?  
(A) 0  
(B) 1  
(C)  $-1$   
(D) Undefined  
(E) Not enough information
- Q5.** Which equation has an undefined slope?  
(A)  $y = 2x + 1$   
(B)  $x = 3$   
(C)  $y = -x + 2$   
(D)  $x + y = 4$   
(E)  $y = x^2$
- Q6.** The slope of a line is  $-3$ . What is its rate of change?  
(A)  $-3$   
(B) 3  
(C) 0  
(D)  $1/3$   
(E) Undefined
- Q7.** A line passes through points  $(2, 3)$  and  $(4, 3)$ . What is its slope?  
(A) 0  
(B) 1  
(C)  $-1$   
(D)  $1/2$   
(E) Undefined
- Q8.** Find the slope of the line  $x = -2y + 1$ .  
(A)  $-1/2$   
(B)  $1/2$   
(C) 0  
(D)  $-2$   
(E)  $1/(-2)$

#### Open Ended Questions (FRQ)

- Q9.** Find the equation of the line that passes through  $(1, 2)$  with a slope of  $-2$ . Show all work.
- Q10.** The cost of a monthly cell phone plan is \$25 plus \$0.05 per text message. Find the rate of change of cost with respect to the number of text messages and interpret its meaning. Show all work.

#### Chef's Tips

- Tip 1: Encourage students to show all work and explain their reasoning for FRQs.
- Tip 2: Use a calculator only when necessary, as most calculations should be done by hand.
- Tip 3: Be sure to review the material thoroughly before the test.